

Alpha decay 2

Let our hypothesis be that the nucleus produces alpha particles at an average rate ν which then tunnel with probability $e^{-2\gamma}$. Hence the decay rate λ is

$$\lambda = \nu e^{-2\gamma}$$

The half-life $\tau_{1/2}$ is

$$\tau_{1/2} = \frac{\log 2}{\lambda} = \frac{\log 2}{\nu} e^{2\gamma}$$

We seek to calculate $\tau_{1/2}$ from γ . The problem is that ν is only known empirically so we use the following modified Geiger-Nuttall formula

$$\log \tau_{1/2} = 3.681 \frac{Z}{\sqrt{E}} - 120.8$$

where $\tau_{1/2}$ is seconds and E is MeV.

For the alpha decay process ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th}$ we have

$$Z = 90$$

$$E = 4.27 \text{ MeV}$$

Hence

$$\tau_{1/2} = 1.43 \times 10^{17} \text{ s} = 4.52 \times 10^9 \text{ yr}$$

The actual half-life of ${}^{238}\text{U}$ is

$$4.463 \times 10^9 \text{ yr}$$

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